VIT
Vellore Institute of Technology
CHENNAI

Reg. Number:	22BAI1266

Continuous Assessment Test (Re-CAT) – II NOVEMBER 2025

Programme	:	B.Tech CSE	Semester	:	Fall 2025-2026
Course Code & Course Title	:	BCSE418L - Explainable Artificial Intelligence	Class Number	:	CH2025260102065 CH2025260102067 CH2025260102523
Faculty	:	Dr. VINOTHINI A Dr. POONKODI Dr. POORNIMA S Dr. RADHIKA SELVAMANI	Slot	:	E1 & E2
Duration	:	90 mins	Max. Mark		50

General Instructions: < Use this space to provide additional information such as graph sheet, data book etc.>

- Write only your registration number on the question paper in the box provided and do not write other information
- Only non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec	Description	Marks	CO	BT Level
1		Raman's research team develops a deep learning model to classify chest X-rays into three categories Normal, Pneumonia and Tuberculosis. This model achieves 95% accuracy on a large dataset having labelled images. However, doctors are doubtful about whether the model is truly recognizing disease regions or simply relying on irrelevant indications such as text markers, patient posture, or image borders. To build trust and interpretability, the team decides to apply Grad-CAM. a)Describe how Grad-CAM can be used on a misclassified pneumonia image to determine whether the model focused on the correct lung regions.(5 marks) b) Suppose the Grad-CAM heatmap shows strong activation around image labels and edges rather than lung tissue. What does this reveal about the model's learning process? Suggest two strategies to correct this issue and improve model trustworthiness.(5 marks)	10	3	3
2		Consider a machine learning model A trained to predict the risk of diabetes in patients. The model is trained on patient data with the following features: Age, Body Mass Index (BMI), Fasting Blood Sugar Level (mg/dL), Physical Activity Level (hours/week), Family History of Diabetes (binary: 1 = Yes, 0 = No).The data science team wants to understand which factors most strongly influence the model's prediction and how each feature affects diabetes risk. (a) Explain in detail how the Accumulated Local Effects (ALE) technique helps visualize the effect of a feature on model predictions.(4 marks) (b) Assume if the ALE plot for BMI shows that diabetes risk increases sharply for BMI values between 25–35 and then levels off beyond 35. What does this suggest about the model's learned	10	2	3

relationship between BMI and diabetes risk? (2 marks)

(c) Suppose the ALE plot for Physical Activity shows almost no change in predicted diabetes risk across all activity levels, but the Leave-One-Feature-Out (LOFO) importance for this feature is 5.8%. Provide two possible reasons for this inconsistency between the ALE and LOFO results. (4 marks)

3

A retail company uses a black-box model to predict customer churn (Yes-1,No-0) based on features such as monthly charges (MC) in dollars, contract type(CT) (Monthly-0, Annual -1) , and tenure(T) in months ,Number of Customer_Support_Call(CS)

(i) How does Local and Global Model-Agnostic XAI methods help interpret the model's predictions in this scenario? Give one example technique for each type.(2 marks)

(ii) Consider the given dataset. Apply the BETA framework to approximate the black-box model with interpretable decision rules. Consider (CT, T) as the neighborhood descriptors and use (MC, CS) to define the decision logic rules within each neighborhood. Construct three decision rule sets and specify two default rules .(8 marks)

C-ID	CT	T	MC	CS	Churn
C001	0	5	95	6	1
C002	1	48	45	1	0
C003	0	8	110	5	1
C004	1	60	55	0	0
C005	0	12	80	4	1
C006	1	30	65	2	0
C007	0	3	100	7	1
C008	1	72	40	1	0

10 | 4 | 3

4

A new company runs a telecom service and wants to reduce customer churn. A Random Forest model is already trained to predict whether a customer is likely to leave. For the customer below, the model predicts a high churn probability of 0.86.

Feature	Value
Monthly_Charges	₹1450
Tenure (Months)	3
Contract_Type	Month-to-Month
Support_Tickets_Last_6M	12
Internet_Usage (GB/Month)	87
Payment_Method	Online Payment

To understand *why* the prediction is high, your analytics team decides to use LIME and SHAP, two Explainable AI methods.

(a) Briefly describe how LIME explains a single prediction.

10 | 3 | 3

LIME gives the following influence scores for this customer:

Feature	Influence on Churn (Positive = increases risk)
Tenure = 3 months	+0.28
Support Tickets = 12	+0.22
Contract Type	+0.18
Monthly Charges	+0.12
Internet Usage	–0.10
Payment Method	–0.04

Interpret and tell the company about why this customer is likely to leave and what might help retain them? **(5 Marks)**

(b) How is SHAP different from LIME in terms of the idea behind how contributions are calculated? SHAP also shows that low tenure and high support ticket count strongly push churn probability up. Explain *why* these two factors might logically increase churn risk, using common-sense customer behavior reasoning.
(5 Marks)

5

A retail chain company is planning to expand and want to decide whether partnering with nearby Mall A or Mall B will increase their customer visits. A survey was conducted in two city zones: North and South. Partners are chosen based on their responses to either partner with Mall 1 or Mall 2 or both. Predicted customer visit counts under different assumptions are shown below:

Partner Mall A	Partner Mall B	Zone	Predicted Customers
yes	yes	North	420
no	yes	North	300
yes	no	North	250
no	no	North	180
yes	yes	South	500
no	yes	South	380
yes	no	South	340
no	no	South	200

a) Compute the Shapley values for the North Zone instance where ShopPlus partners with Mall A only (i.e., Partner A = yes, Partner B = no) and draw a SHAP plot. **(6 marks)**

b) Based on the SHAP interpretation, recommend whether the company should partner with Mall A, Mall B, or both to maximize customer visits. Justify your answer. **(4 marks)**

10 | 3 | 3

**********All the best **********